



AI Workflow & ReAct Agent

Week 5

Last Week

Week 4: Tools for building AI Agents

LangChain, Azure Open AI Api, LLMod.ai

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Applications that can reason. Powered by LangChain.

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Text Generation

What it is: Generate Text from Prompt

Course Model: gpt-5-mini

Cost per 1M tokens:

Text tokens

Per 1M tokens • Batch API price ☐

Input

\$0.25

Cached input

\$0.025

Output

\$2.00

LLMod.ai

What it is: System to use models, manage api keys & track budgets

The screenshot displays the LLMMod.ai Instructor Console interface. The top navigation bar includes the LLMMod.ai logo and 'Instructor Console' text. A breadcrumb trail shows 'Test Course 1 > Assignments > 1'. The left sidebar lists 'Test Course 1' and 'Assignment 1'. The main content area is titled 'Assignment 1' and 'Individual Assignment'. It features three summary cards: 'Assignment Period' (11/25/2025 to 12/2/2025), 'Budget' (\$5.00), and 'Cost Used' (\$0.0000 with a progress bar indicating 0.0% used). Below these is an 'API Key' section with the text 'Your personal API key for this assignment', a 'Current Key' field showing 'sk-...G9DQ', and a 'Base URL' field showing 'https://api.llmod.ai'. A 'Rotate Key' button is located to the right of the current key field.

LLMod.ai
Instructor Console

Test Course 1 > Assignments > 1

Test Course 1

Assignment 1

Assignment 1

Individual Assignment

Assignment Period

11/25/2025
to 12/2/2025

Budget

\$5.00

Cost Used

\$0.0000

0.0% used

API Key
Your personal API key for this assignment

Current Key: sk-...G9DQ

Base URL: https://api.llmod.ai

Rotate Key

This Week

Week 4: Tools for building AI Agents

LangChain, Azure Open AI Api, LLMMod.ai

Week 5: AI Workflow & ReAct Agent

AI Workflow, ReAct Agent, Individual Project Launch

Example – Toby HR

HR departments are overwhelmed

Problem Statement

📁 HR departments receive a high volume of resumes for job openings, making it challenging to efficiently and fairly evaluate candidates.

🕒 Manual sorting is time-consuming and prone to human biases.

🔄 Communication with candidates is also time consuming and difficult.



Sorting Resumes (HR system)

Name:

Toby, Auto HR

One liner:

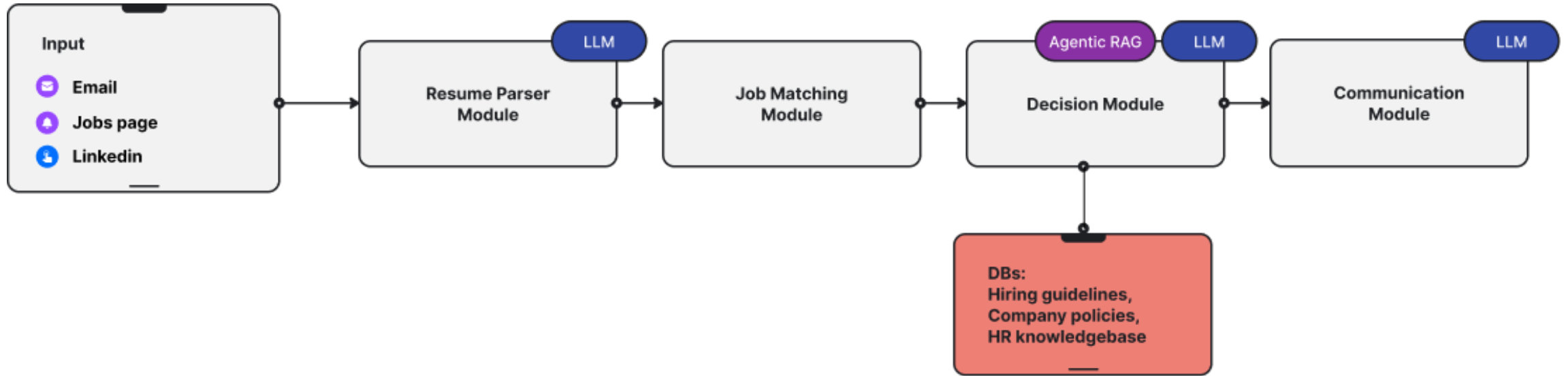
Automates routine HR tasks such as sorting resumes, matching candidates to job openings, and managing candidate communication to streamline recruitment.



We want a system that can:

1. Sort qualified resume according to open positions
2. Identify and prioritize candidates for initial interviews based on compatibility scores and role urgency.
3. Communicate with candidates before first interview.

Toby, Auto HR



Resume Parser Module

Purpose:

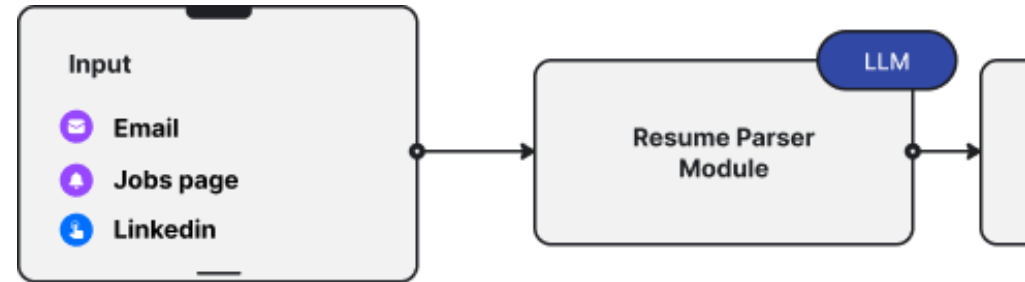
Determine how well a candidate fits each open position

What It Does:

1. Extracts skills, experience, education, tools, and work history.
2. Resolves inconsistent wording ("ML Engineer" vs "Machine Learning Engineer").

How it Works:

1. Parse resumes documents.
2. Use LLM to convert extracted text to strict JSON.



Job Matching Module

Purpose:

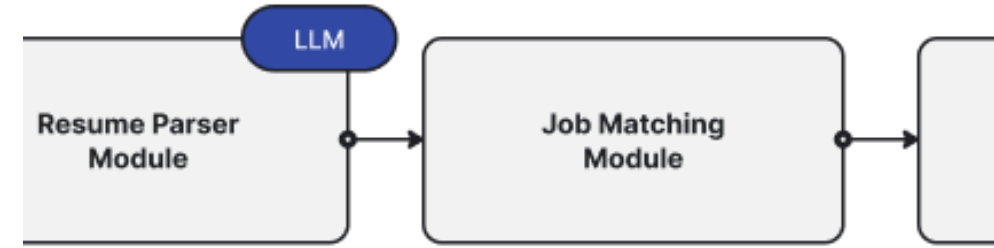
Convert messy CVs into structured, machine-readable data

What It Does:

1. Checks must-have requirements.
2. Calculates a match score (0–100).
3. Produce (candidate, rule) ranking list

How it Works:

1. Deterministic rules for: location, seniority, years of experience.
2. Optional LLM: semantic skill similarity (“TensorFlow” ≈ “Deep Learning”).



Decision Module

Purpose:

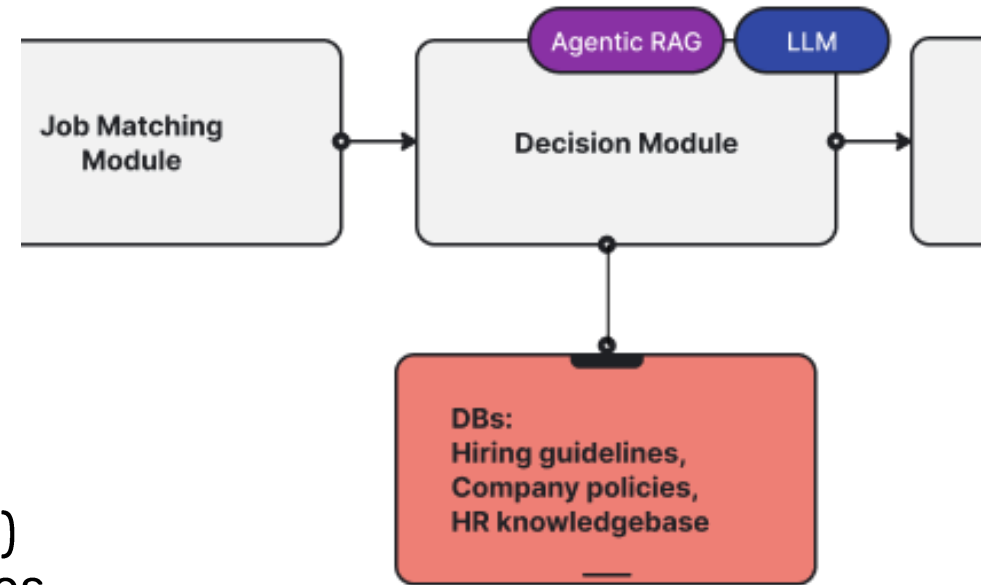
Make final decision move forward to first interview based on rules, scores, and company knowledge & policies

What It Does:

1. Determines: interview, reject.
2. Handles ambiguous cases (i.e. time sensitive positions)
3. Applies internal hiring guidelines (policies, fairness rules, diversity constraints).

How it Works:

1. Retrieves relevant documents (RAG).
2. LLM uses reason and thinking to provides human like decisions.



Communication Module

Purpose:

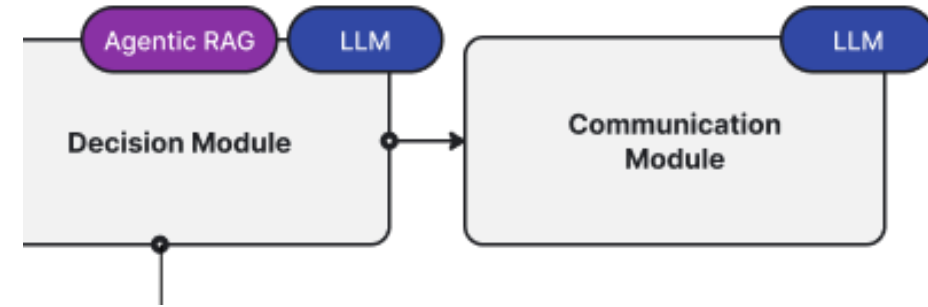
Automatically generate professional, personalized messages

What It Does:

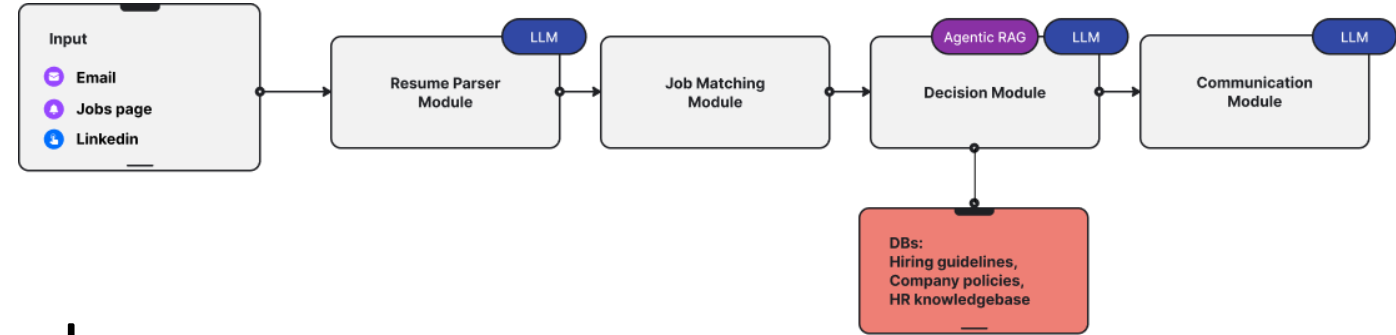
1. Creates personalized interview invitations, rejections.
2. (Optional) Can answer some candidates questions regarding rejections

How it Works:

1. LLM Creates personalized emails according to decision and context about the candidate.



Am I an Agent?



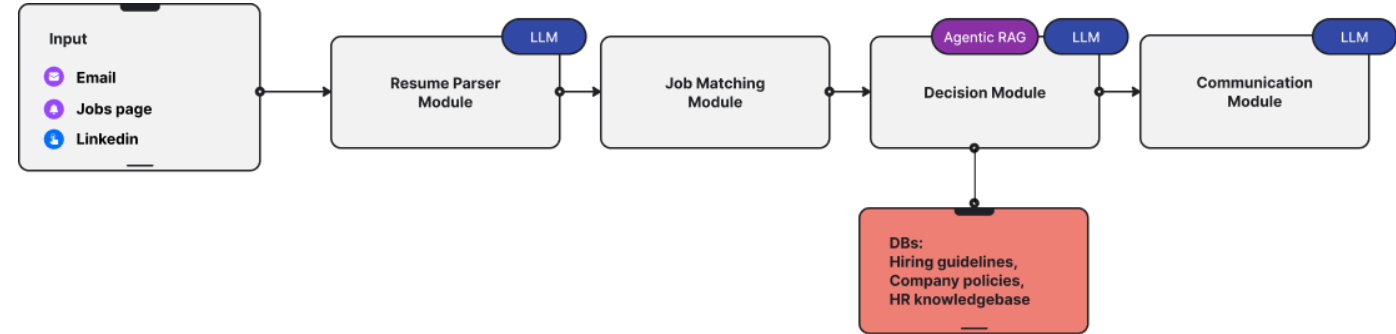
✓ Perceive its environment

✓ Reason

✗ Act Autonomously

Automation != Autonomously

AI Workflow



What it is:

An AI Workflow/Pipeline is a structured, step-by-step pipeline where data flows through a series of predefined modules, and each module performs a specific task.

In an AI Workflow

- The steps are fixed.
- Each module takes an input, transforms it, and passes it forward.
- The system does not make decisions about what to do next.

* LLMs may be used inside the steps, but the overall flow is still fixed

AI Workflow - Pros

- ✓ Simple and reliable
- ✓ Cheap to run
- ✓ Easy to understand and debug
- ✓ Easy to scale
- ✓ Great for some tasks

Questions

What if a candidate matches multiple positions?

What if the resume is unclear/incomplete and we need to ask a follow-up question?

What happens if position filled and there are candidates in the pipeline?

Does the AI Workflow support this?

HR still overwhelmed



Let's try something else

Tools

Resume Parser

Job Matching

Decision Module

Communication

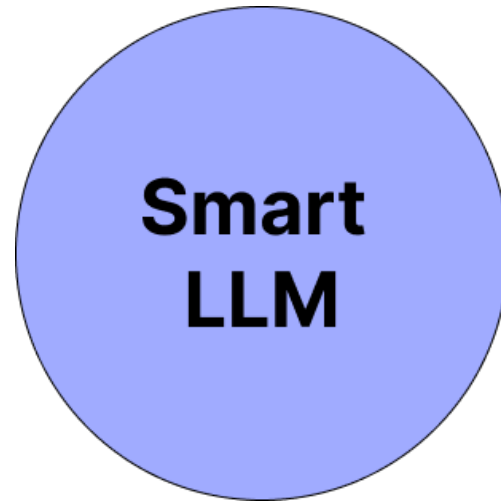
Email Connectivity

Calendar Connectivity

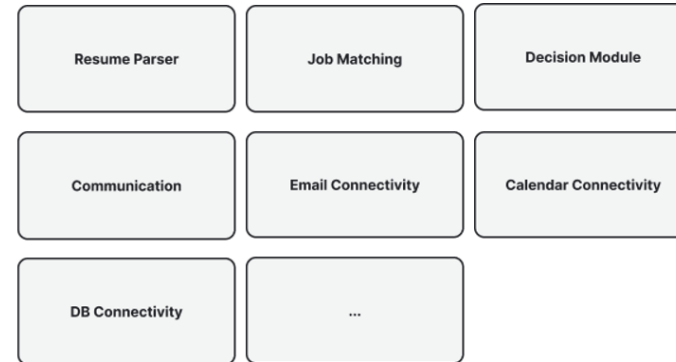
DB Connectivity

...





Tools



* Good Reasoning Capabilities

Give the LLM tasks

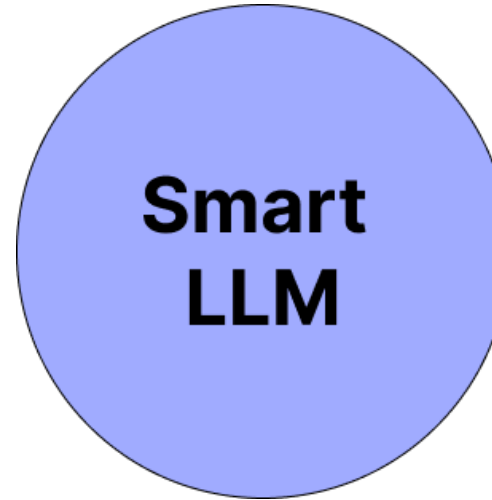
Make hiring Decision,
schedule an interview or
reject this candidate
[cv.pdf]

Candidate X want to
reschedule interview

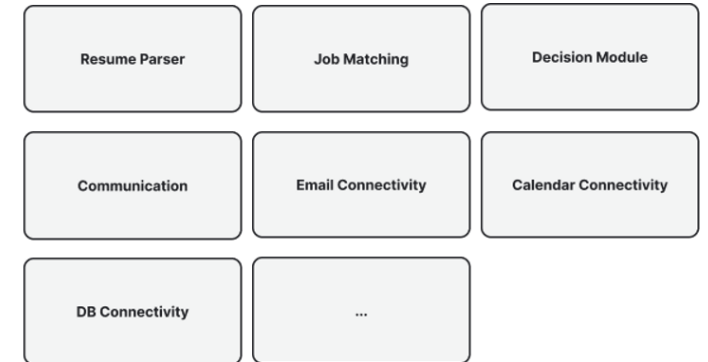
Position Y filled, cancel any
outstanding interview and
route candidates to
different position if
applicable

...

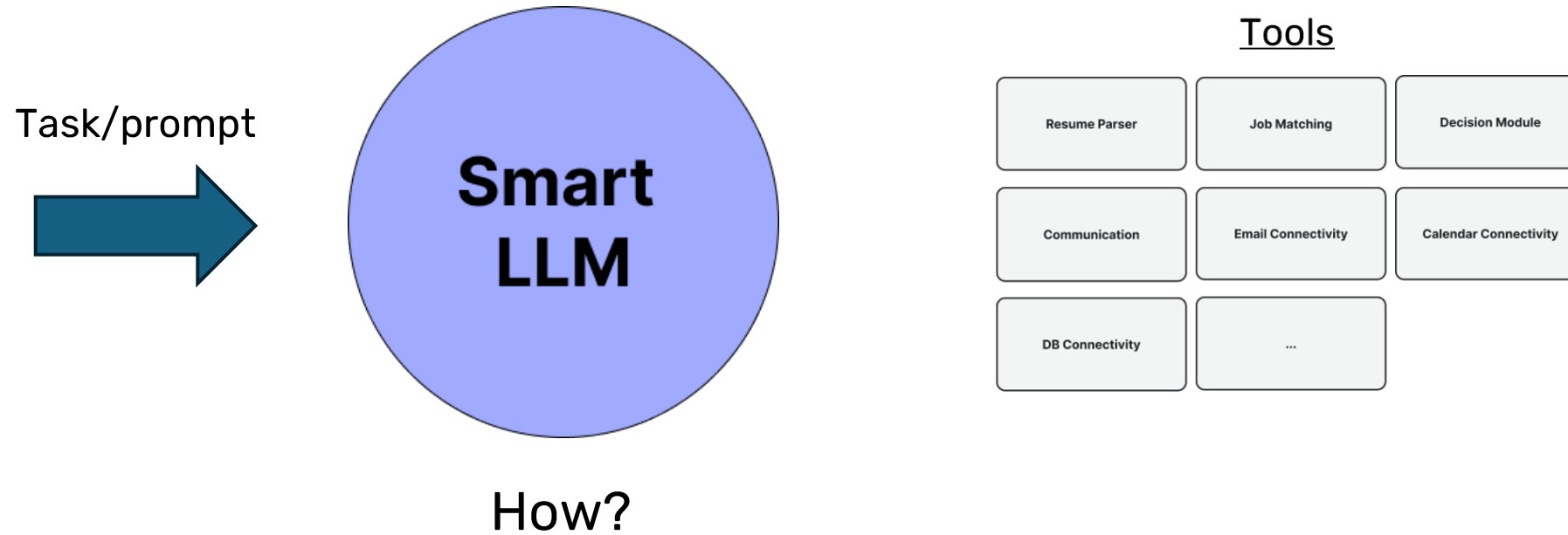
Task/prompt



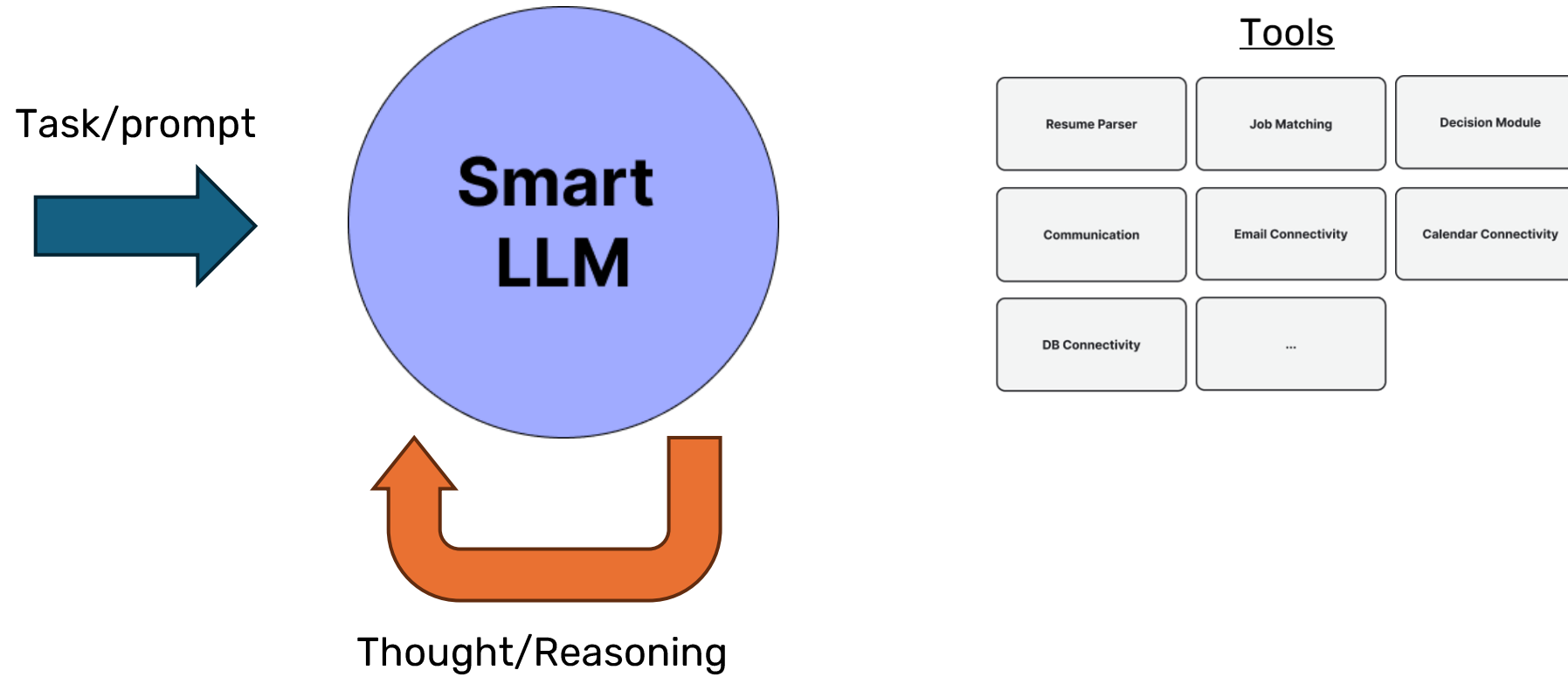
Tools



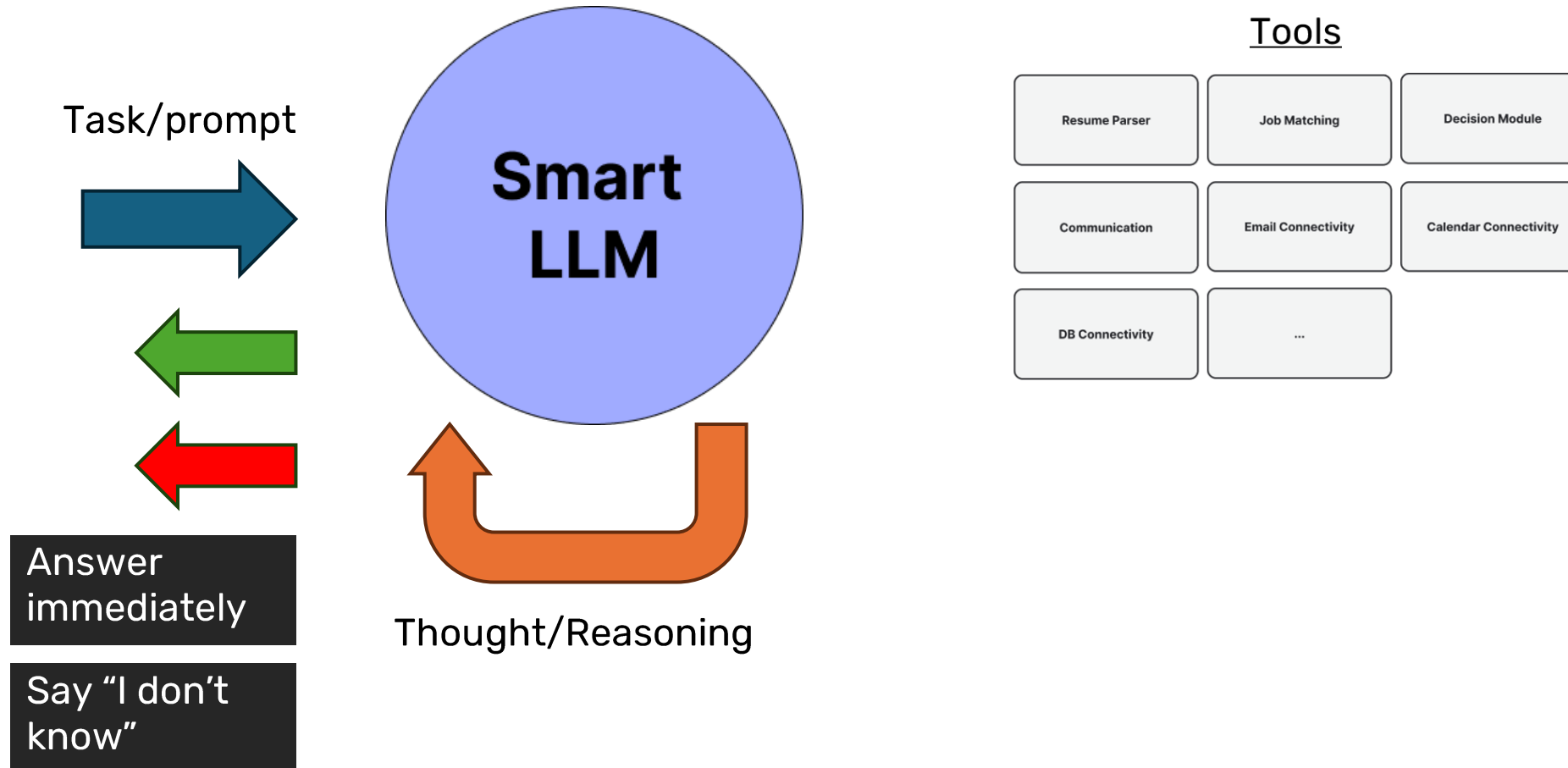
LLM Decide what to do



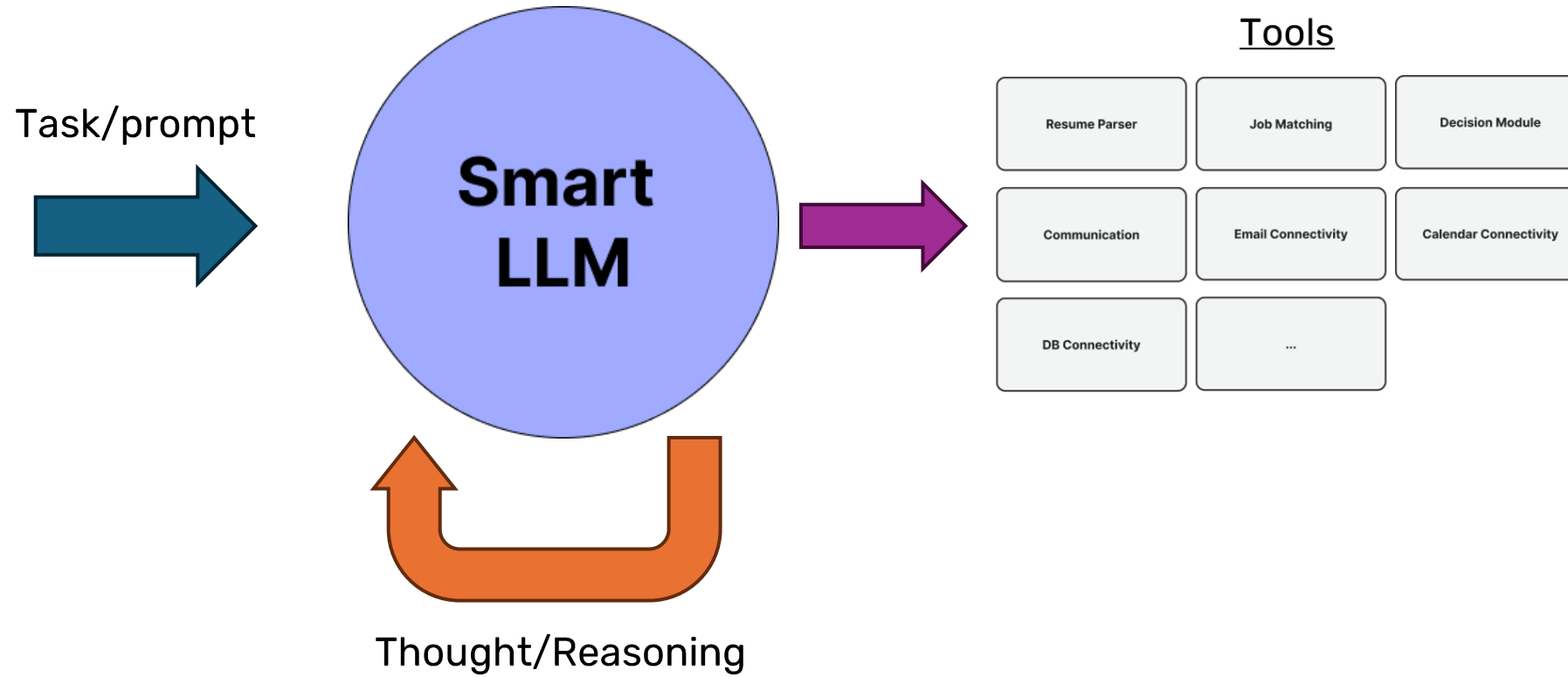
LLM Decide what to do



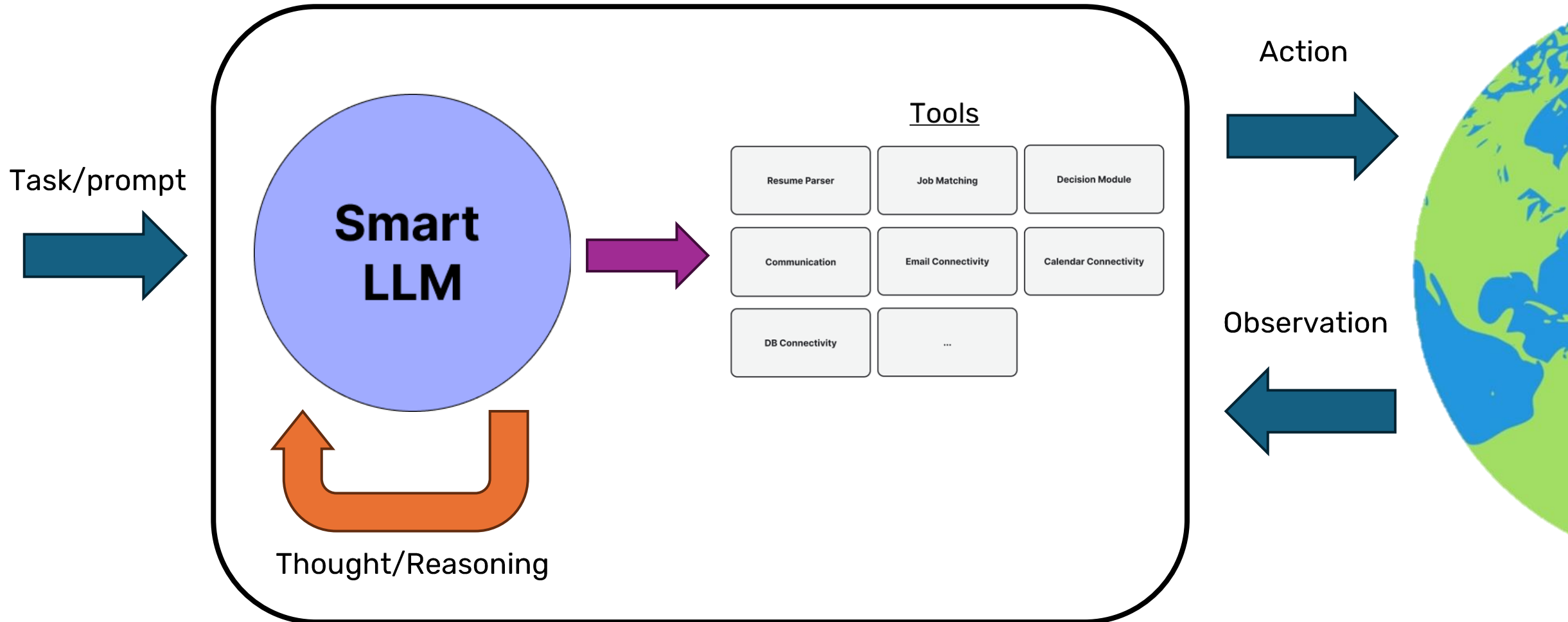
What can the LLM do?



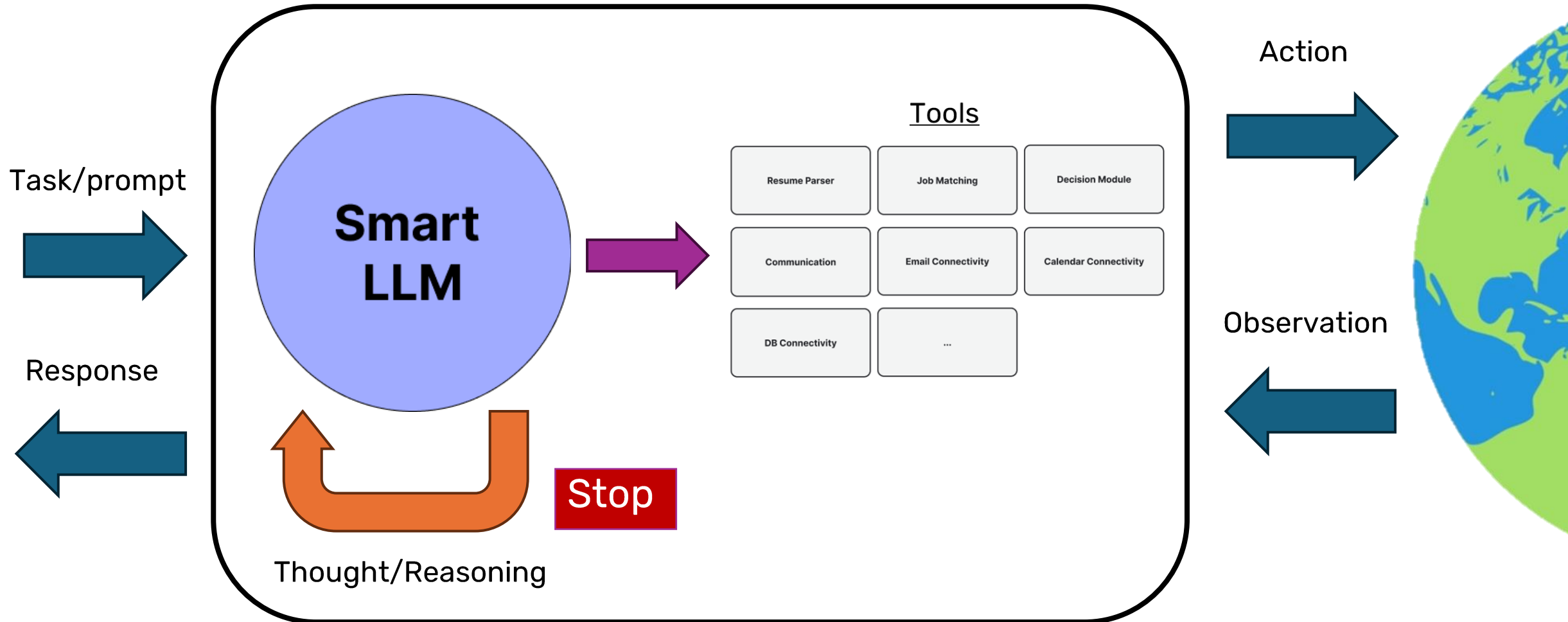
What can the LLM do?



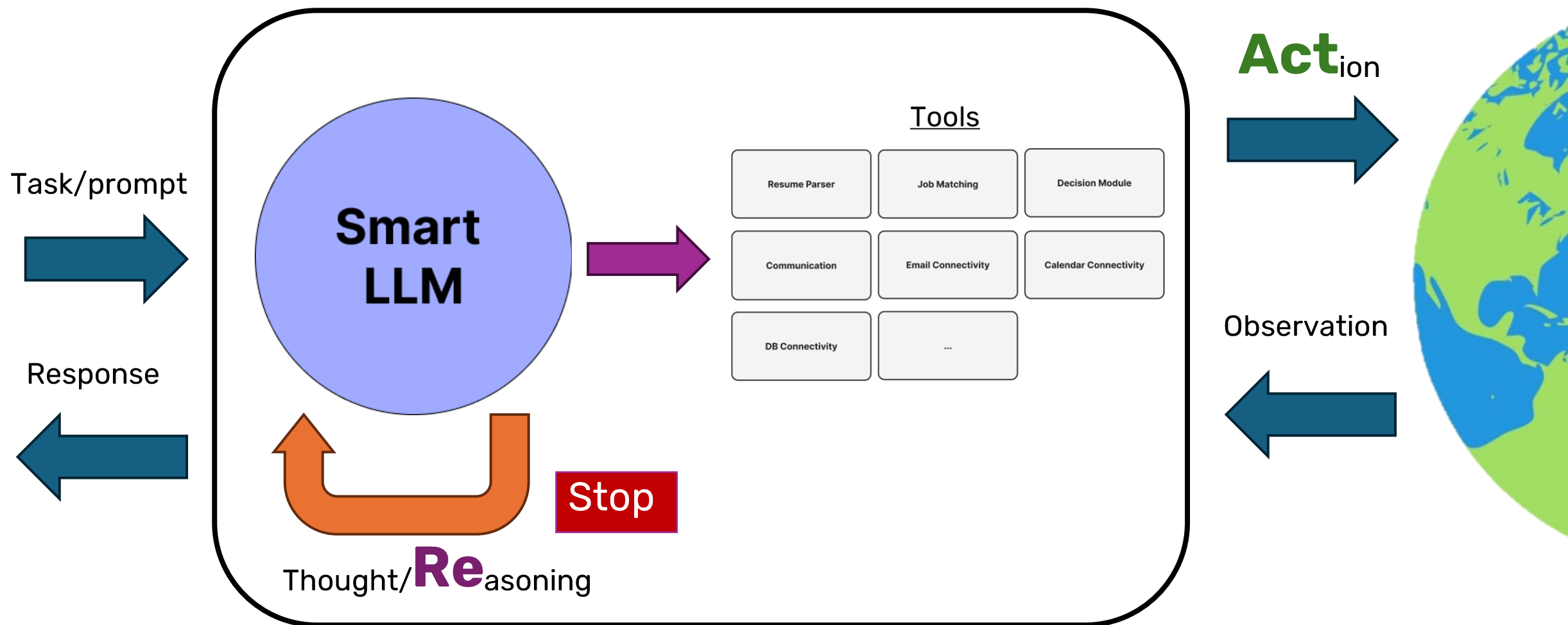
Action And Observation



What's missing? ∞



ReAct Agent



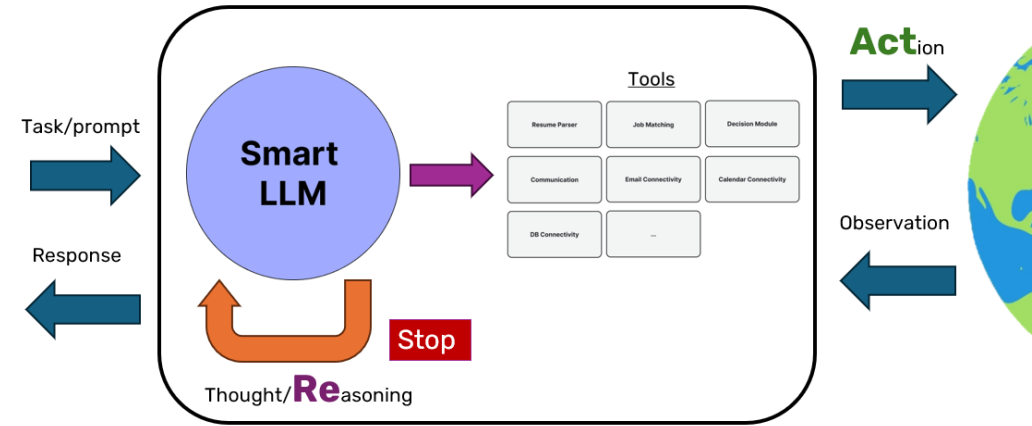
ReAct Agent

What it is:

A ReAct Agent is an AI system that can both reason and act. It uses a loop of Thought -> Action -> Observation to decide what to do next, instead of following a fixed workflow.

ReAct Agent Can:

- **Think** through a problem step-by-step
- **Choose actions** (like calling tools, APIs, search, or RAG)
- **Observe results** and update its plan
- **Ask clarifying questions** when information is missing
- **Adapt the workflow** based on the situation
- **Handle multi-step, messy, real-world tasks**



Assignment 2

Due date: 14.6.2026 End Of Day

Weight: Part of Course Project

Assignment 2: Find Data Sources for your AI Agents

Goal

For 2 domains/problems out of the 3 you've submitted, identify the data sources that will serve as the context store for your AI agent.

Instructions

Identify data sources for each of your domains/problems. These may include:

- **RAG-suitable sources**, and
- **regular data sources** the agent can query directly.

You may use one or multiple data sources per problem.

Each domain must include **at least one RAG-suitable source**.

The **total data size per problem/domain** (combined across all sources) can't exceed **50MB**.

Submission Format

Add the relevant data-source details to 2 out of 3 of your **Assignment 1 submission**, at the end of each domain/pain-point section, and **resubmit the form**.

Project meetings:

Batch 1: 15.6.2026 (9:30 – 11:20)
Batch 2: 22.6.2026 (9:30 – 11:20)
Batch 3: 29.6.2026 (9:30 – 11:20)

סטודנט\ית 3	סטודנט\ית 2	סטודנט\ית 1	שם קבוצה	שעת מפגש	תאריך מפגש
				09:30	15.6.2026
				09:40	15.6.2026
				09:50	15.6.2026
				10:00	15.6.2026
				10:10	15.6.2026
				10:20	15.6.2026
				10:30	15.6.2026
				10:40	15.6.2026
				10:50	15.6.2026
				11:00	15.6.2026
				11:10	15.6.2026
				09:30	22.6.2026
				09:40	22.6.2026
				09:50	22.6.2026
				10:00	22.6.2026
				10:10	22.6.2026
				10:20	22.6.2026
				10:30	22.6.2026
				10:40	22.6.2026
				10:50	22.6.2026
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				11:10	22.6.2026
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				09:40	29.6.2026
				09:50	29.6.2026
				10:00	29.6.2026
				10:10	29.6.2026
				10:20	29.6.2026
				10:30	29.6.2026
				10:40	29.6.2026
				10:50	29.6.2026

Questions?